INCH-POUND

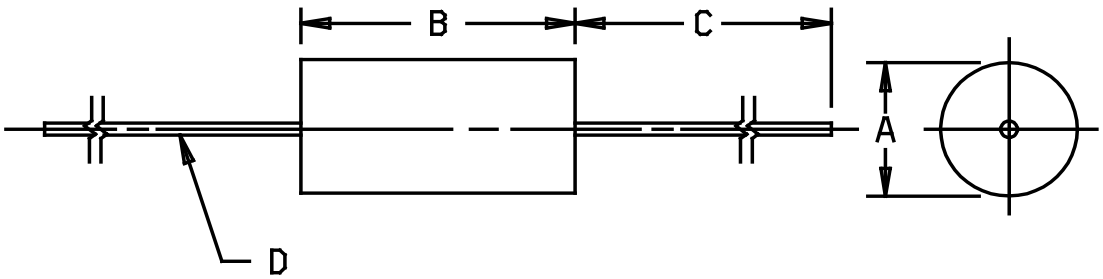
MIL-PRF-39010/4E
W/AMENDMENT 2
14 December 2022
SUPERSEDING
MIL-PRF-39010/4E
W/AMENDMENT 1
6 September 2018

PERFORMANCE SPECIFICATION SHEET

COILS, FIXED, RADIO FREQUENCY, MOLDED, SUBMINIATURE (PHENOLIC CORE), ESTABLISHED RELIABILITY AND NONESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39010.



Letter	Dimensions are in inches with metric equivalents (mm) in parentheses	Dimensions are in inches with metric equivalents (mm) in parentheses
	Minimum	Maximum
A	.157 (3.99)	.219 (5.56)
B	.407 (10.34)	.469 (11.91)
C	1.250 (31.75)	1.626 (41.30)
D	.023 (0.58)	.027 (0.69)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 1. Dimensions and configuration.

AMSC N/A FSC 5950

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REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Material: Phenolic core.

Weight: 0.03 ounce maximum.

Operating temperature range: -55°C to +125°C.

Dielectric withstanding voltage: MIL-STD-202-301; test voltage of 1,000 V rms.

Barometric pressure: MIL-STD-202-105, test condition C, 70,000 feet with a test voltage of 200 V rms.

Electrical characteristics: See table I and table II.

Inductance: See table I.

Inductance tolerance: See table I.

Q values: See table I.

Self-resonant frequency: See table I.

DC resistance: See table I.

Temperature rise: 35°C.

Terminal pull: 5 pounds.

Part or Identifying Number (PIN): M39010/04 (dash number from table I).

Supersession data: This specification supersedes MS75008.

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TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance μH	Inductance tolerance ± percent	Q minimum	Test frequency (MHz)	Self-resonant frequency minimum (MHz)	DC resistance (25°C) maximum (ohms)	Rated dc current (mA) 2/
BR15**	.15	5, 10, 20	55	25.0	510	0.030	2,900
BR16**	.16	5	50	25.0	415	0.035	2,700
BR18**	.18	5, 10	50	25.0	415	0.035	2,700
BR20**	.20	5	50	25.0	415	0.035	2,700
BR22**	.22	5, 10, 20	50	25.0	415	0.035	2,700
BR24**	.24	5	50	25.0	350	0.065	2,000
BR27**	.27	5, 10	50	25.0	350	0.065	2,000
BR30**	.30	5	50	25.0	350	0.065	2,000
BR33**	.33	5, 10, 20	50	25.0	350	0.065	2,000
BR36**	.36	5	50	25.0	300	0.085	1,700
BR39**	.39	5, 10	50	25.0	300	0.085	1,700
BR43**	.43	5	50	25.0	300	0.085	1,700
BR47**	.47	5, 10, 20	50	25.0	300	0.085	1,700
BR51**	.51	5	50	25.0	270	0.125	1,450
BR56**	.56	5, 10	50	25.0	270	0.125	1,450
BR62**	.62	5	45	25.0	250	0.150	1,300
BR68**	.68	5, 10	45	25.0	250	0.150	1,300
BR75**	.75	5	40	25.0	210	0.205	1,100
BR82**	.82	5, 10	40	25.0	210	0.205	1,100
BR91**	.91	5	40	25.0	200	0.290	930
B1R0**	1.00	5, 10	40	25.0	200	0.290	930
B1R1**	1.10	5	30	7.9	180	0.400	785
B1R2**	1.20	5, 10	30	7.9	180	0.400	785
B1R3**	1.30	5	30	7.9	170	0.485	720
B1R5**	1.50	5, 10	30	7.9	170	0.485	720
B1R6**	1.60	5	30	7.9	150	0.740	580
B1R8**	1.80	5, 10	30	7.9	150	0.740	580
B2R0**	2.00	5	30	7.9	140	0.970	505
B2R2**	2.20	5, 10	30	7.9	140	0.970	505
B2R4**	2.40	5	30	7.9	120	1.20	460
B2R7**	2.70	5, 10	30	7.9	120	1.20	460

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the product level (e.g., C, M, P, R, S) and will be added to the end of the dash number.
2/ The rated dc current is based on 90°C ambient temperature with a 35°C rise.

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TABLE II. Electrical characteristics (final). 1/

Inspection group	Allowable variation from initial measurement	Allowable variation from initial measurement	Allowable percent from specified minimum value in electrical characteristics (initial) table	Allowable percent from specified minimum value in electrical characteristics (initial) table
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±2			-10
Group IV	±5	±(3% +.001 ohm)	2/	-10
Group V	±5	±(2% +.001 ohm)	2/	-15
Quality conformance inspection Group B				
Subgroup 1	±5	±(2% +.001ohm)	2/	-15
Subgroup 3	±2			-10
Subgroup 4	±5	±(3% +.001 ohm)	2/	-10

1/ Test fixture allowance of +.01 μH shall be added to all change in inductance limits ±(_ percent +.01 μH).
2/ The self-resonant frequency shall be not less than the value specified in table I.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-39010, this document references the following:

MIL-STD-202-105, MIL-STD-202-301

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Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

Review activities:
Army - AR, CR4, MI
Navy - AS, CG, MC, OS, SH
Air Force - 19

(Project 5950-2022-072)

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